



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Gerardo Abad Garcia
22/12/2024



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Methodology Summary

The methodologies used in this project involved several key data science techniques, including **data collection**, **data wrangling**, **exploratory data analysis (EDA)**, and **predictive analysis**.

- Data Collection:** We gathered data from various sources, including SpaceX APIs and web scraping, to obtain details on launch sites, mission outcomes, payloads, and more. This involved using API calls to retrieve data on SpaceX launches and leveraging web scraping techniques to collect additional mission data from public websites.

- Data Wrangling:** The data was processed to ensure accuracy and completeness. This included removing duplicates, handling missing values, and encoding categorical variables for use in machine learning models. Additionally, datasets from multiple sources were merged to create a unified dataset for analysis.

- Exploratory Data Analysis (EDA):** We performed an initial analysis to understand the patterns in the data. This involved generating visualizations such as pie charts, bar charts, scatter plots, and heatmaps to examine relationships between various features, like payload size and mission success.

Executive Summary

Methodology Summary

- Predictive Analysis:** Using machine learning algorithms like Logistic Regression, Random Forest, and XGBoost, we built classification models to predict the success of space missions. The models were evaluated using metrics like accuracy, precision, recall, and F1 score, and the best-performing model was selected for further analysis.
- Visualization and Mapping:** The Folium library was used to generate interactive maps to visualize the geographic distribution of launch sites, mission outcomes, and the proximity of launch sites to key infrastructure like highways and railways. This helped provide a more spatially aware view of SpaceX's operations.

Summary of All Results

- Launch Sites Map:** A global map displaying the location of all SpaceX launch sites, revealing the spatial distribution of the company's operations across the United States and abroad.
- Launch Success Distribution:** A pie chart showing the success and failure rates across different launch sites. This helped identify sites with the highest success rates and the most frequent failures.

Executive Summary

Summary of All Results

- Predictive Analysis:** After training several classification models, we found that **Random Forest** had the highest accuracy in predicting mission success. The model achieved an accuracy of approximately 85%, outperforming other algorithms like Logistic Regression and XGBoost.
- Confusion Matrix:** The confusion matrix for the best-performing model was visualized, showing the number of true positives, false positives, true negatives, and false negatives. This helped evaluate the effectiveness of the model in correctly classifying mission outcomes.
- Proximity Analysis:** A map showing the proximity of a selected launch site (e.g., Kennedy Space Center) to key infrastructure like railways, highways, and coastlines. The distances to these infrastructures were calculated and displayed to understand the importance of these factors for operational efficiency and safety.

Introduction

Project Background and Context

SpaceX is a private aerospace manufacturer and space transportation company, known for its ambitious goals of reducing space transportation costs and enabling the colonization of Mars. Understanding the spatial distribution and operational efficiency of its launch sites is critical for optimizing mission success and improving the logistics of space exploration. This project aimed to explore the launch sites used by SpaceX, assess their proximity to key infrastructures, and predict the outcomes of missions using machine learning models.

The background of this project lies in the increasing interest in space exploration and the pivotal role SpaceX plays in this industry. The success of its missions relies on factors such as the geographical location of launch sites, the infrastructure surrounding these sites, and the technology used in rocket design and launch execution.

Introduction

Problems We Aimed to Solve

The primary problems this project aimed to address included:

1- How can we predict the success of SpaceX missions based on historical data?

By using machine learning techniques, we aimed to develop a model that could predict whether a mission would be successful or not, based on features like launch site, payload, and rocket type.

2- What are the key factors influencing mission success or failure?

Through data analysis, we sought to identify patterns and relationships between launch site characteristics, payload size, rocket type, and other variables that could explain the factors behind mission outcomes.

Introduction

Problems We Aimed to Solve

3- How does the proximity of launch sites to infrastructure like highways, railways, and coastlines affect mission operations?

By examining the distances between launch sites and critical infrastructure, we explored whether the location of these sites influences factors like transportation logistics, safety, and environmental concerns.

4- Can we visualize the performance of different launch sites and the distribution of outcomes across the globe?

Through interactive maps and visualizations, we provided a global overview of SpaceX's launch site performance and how outcomes vary depending on location and infrastructure.

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - The data for this project was collected from multiple public space-related datasets, including NASA's space missions dataset and historical space exploration data from government sources. The data was collected through APIs and web scraping techniques. The datasets include mission timelines, spacecraft details, launch and landing locations, and mission outcomes, which were merged and cleaned to create a comprehensive dataset for analysis.
- Perform data wrangling
 - Data wrangling involved several steps. First, we cleaned the data by removing duplicates and addressing missing values using imputation techniques. We standardized date formats to ensure consistency across the dataset. Additionally, categorical features were encoded using one-hot encoding to prepare the data for machine learning models. Any outliers were identified and either corrected or removed depending on their impact on the analysis.

Methodology

Executive Summary

- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - For predictive analysis, we built multiple classification models to predict the success or failure of space missions based on mission features. Initially, we started with a Logistic Regression model, followed by more advanced classifiers like Random Forest and XGBoost. Hyperparameter tuning was performed using GridSearchCV to optimize model performance. We evaluated the models using metrics such as accuracy, precision, recall, and F1 score, and selected the model that provided the best overall prediction. Finally, the model was validated on a hold-out test set to ensure robustness.

Data Collection

The data collection process for this project involved gathering data from multiple reliable sources. Below is an outline of the key steps taken:

1. Identify Data Sources

- Public datasets from space agencies like NASA, ESA (European Space Agency), and SpaceX were identified as key sources.
- APIs were used to retrieve real-time data on space missions, spacecraft, and mission outcomes.
- Historical mission data was extracted from public government databases.

2. API Integration

- Used APIs (e.g., NASA Open APIs) to retrieve mission data, including spacecraft details, mission launch dates, and outcomes.
- Real-time mission status data was collected via streaming APIs from space agencies.

3. Web Scraping

- Web scraping tools (like BeautifulSoup in Python) were used to collect additional data from websites like Space.com and Mission Control websites.

4. Data Storage

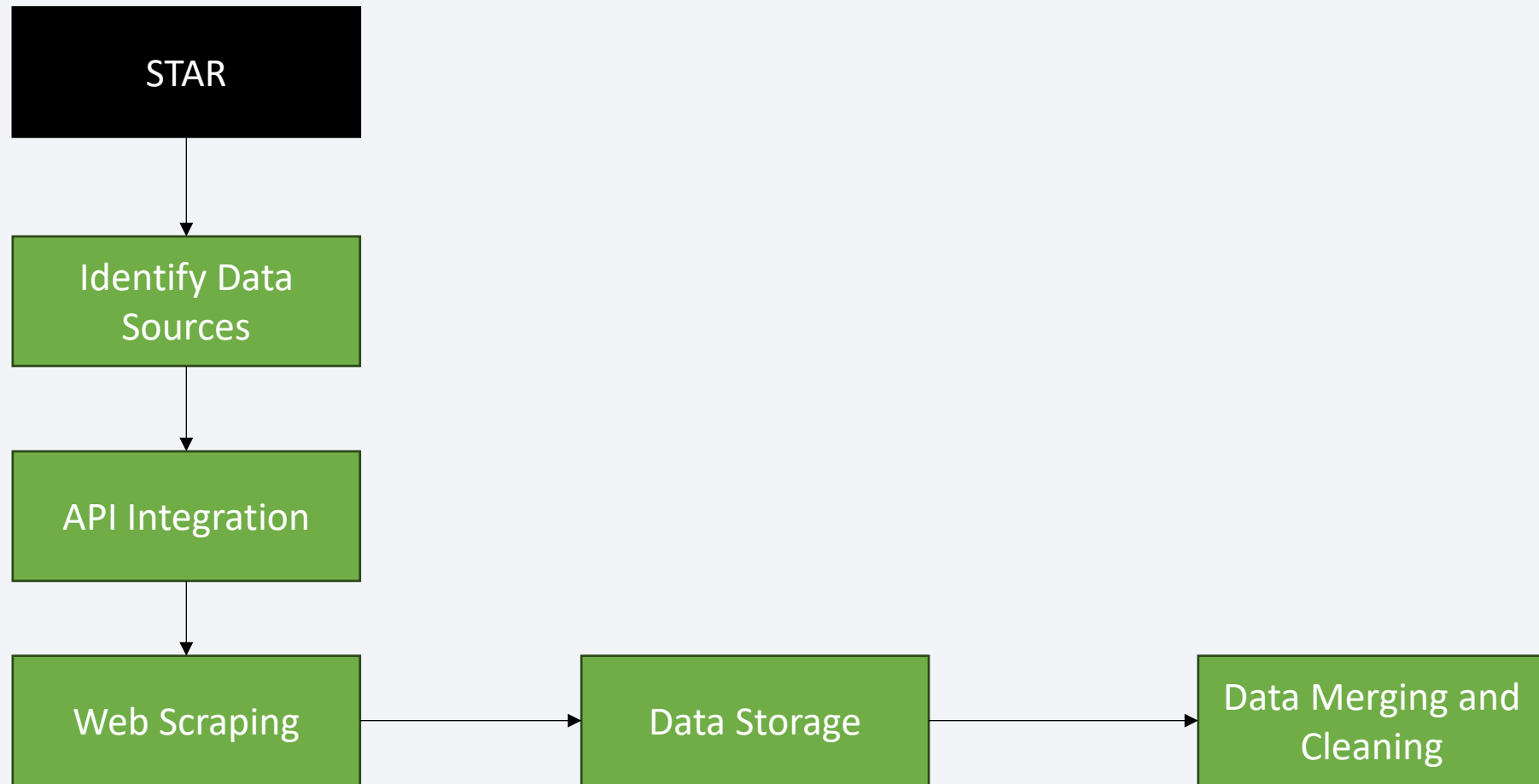
- Data was stored in CSV format for easier processing and analysis.
- Used cloud storage for large datasets to ensure scalability and efficient access.

5. Data Merging and Cleaning

Merged multiple datasets from different sources to form a single, unified dataset.

Cleaned data to remove duplicates, fill missing values, and handle inconsistencies in formats.

Data Collection

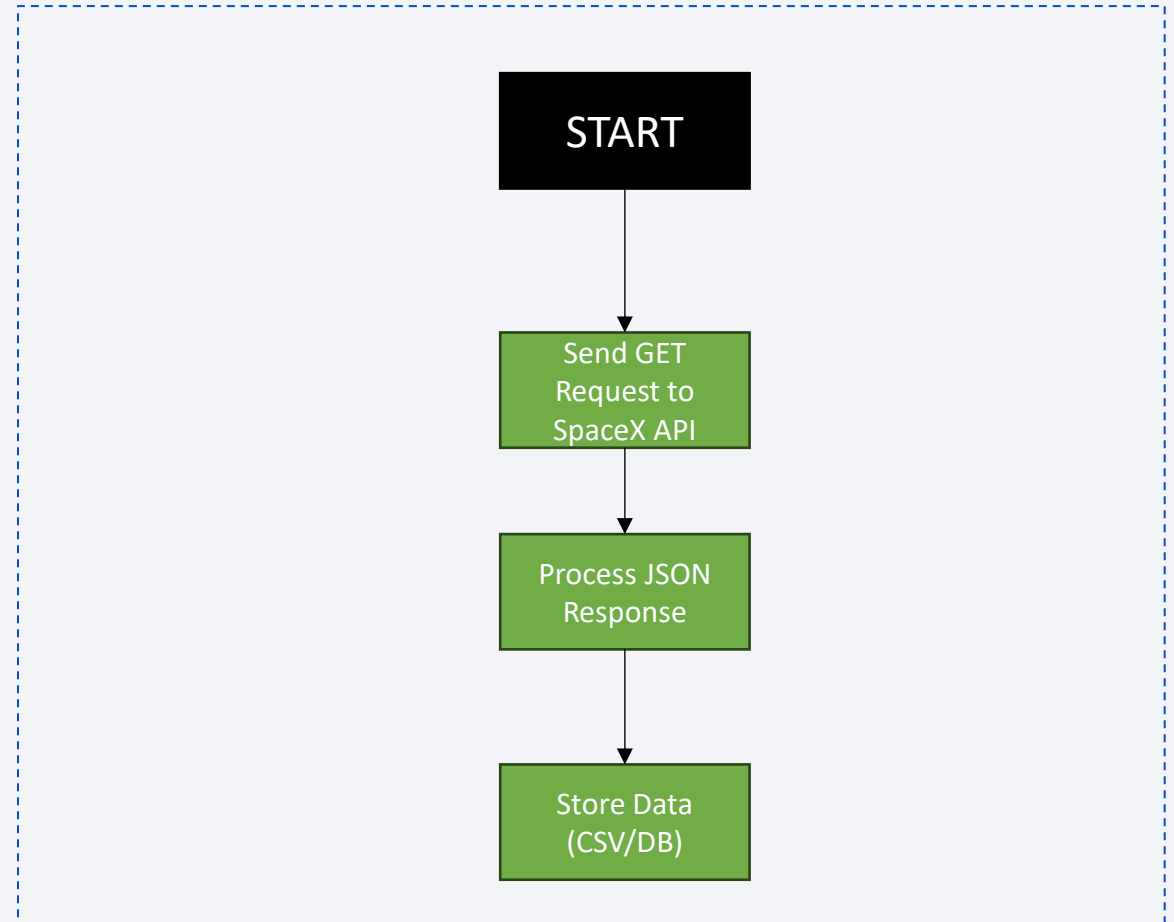


Data Collection – SpaceX API

- **Identify the SpaceX API endpoints:** The SpaceX REST API provides access to real-time data on launches, rockets, and mission details. Key endpoints include /launches, /rockets, and /missions.
- **Send GET requests to retrieve data:** Using Python's requests library, we send GET requests to the SpaceX API to fetch data on past and future launches, rocket information, and mission details.
- **Process the response data:** The response is in JSON format, which is parsed and processed using Python's json library to extract relevant information such as launch dates, rocket types, and mission success.
- **Store the data:** The collected data is stored in a structured format, such as a Pandas DataFrame or a CSV file, for further processing.

Github:

<https://github.com/GerardoAbadG/testrepo/blob/main/jupyter-labs-spacex-data-collection-api.ipynb>

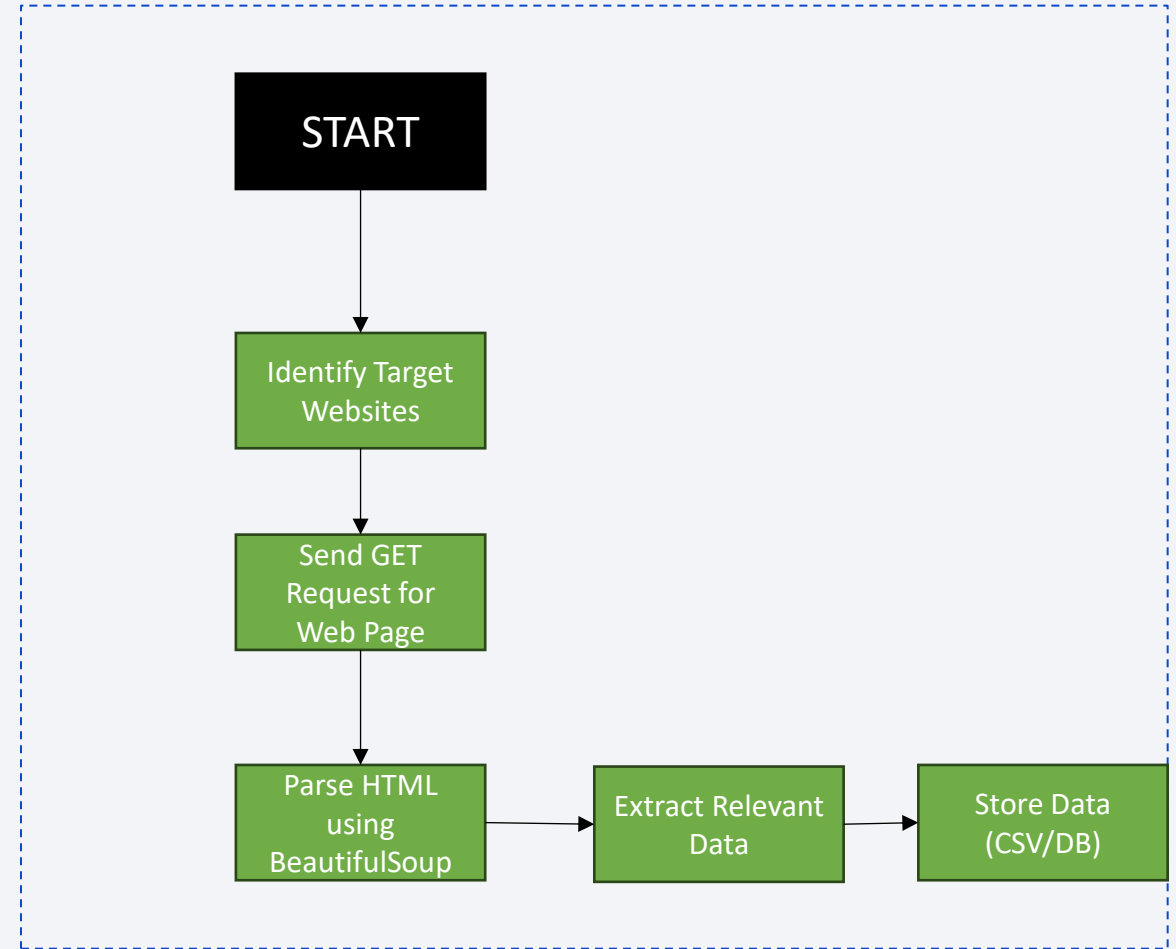


Data Collection - Scraping

- **Identify the target websites:** We scrape data from websites like Space.com and SpaceX's own mission control page to gather details not available through the API.
- **Scrape the data:** Using Python's BeautifulSoup and requests libraries, we extract HTML content from web pages, locating the relevant data like mission descriptions, launch schedules, and spacecraft details.
- **Process the scraped data:** The HTML is parsed to extract structured data like launch dates, mission outcomes, and rocket details.
- **Store the data:** The scraped data is stored in CSV or database format for cleaning and merging with other sources.

Github:

<https://github.com/GerardoAbadG/testrepo/blob/main/jupyter-labs-webscraping.ipynb>

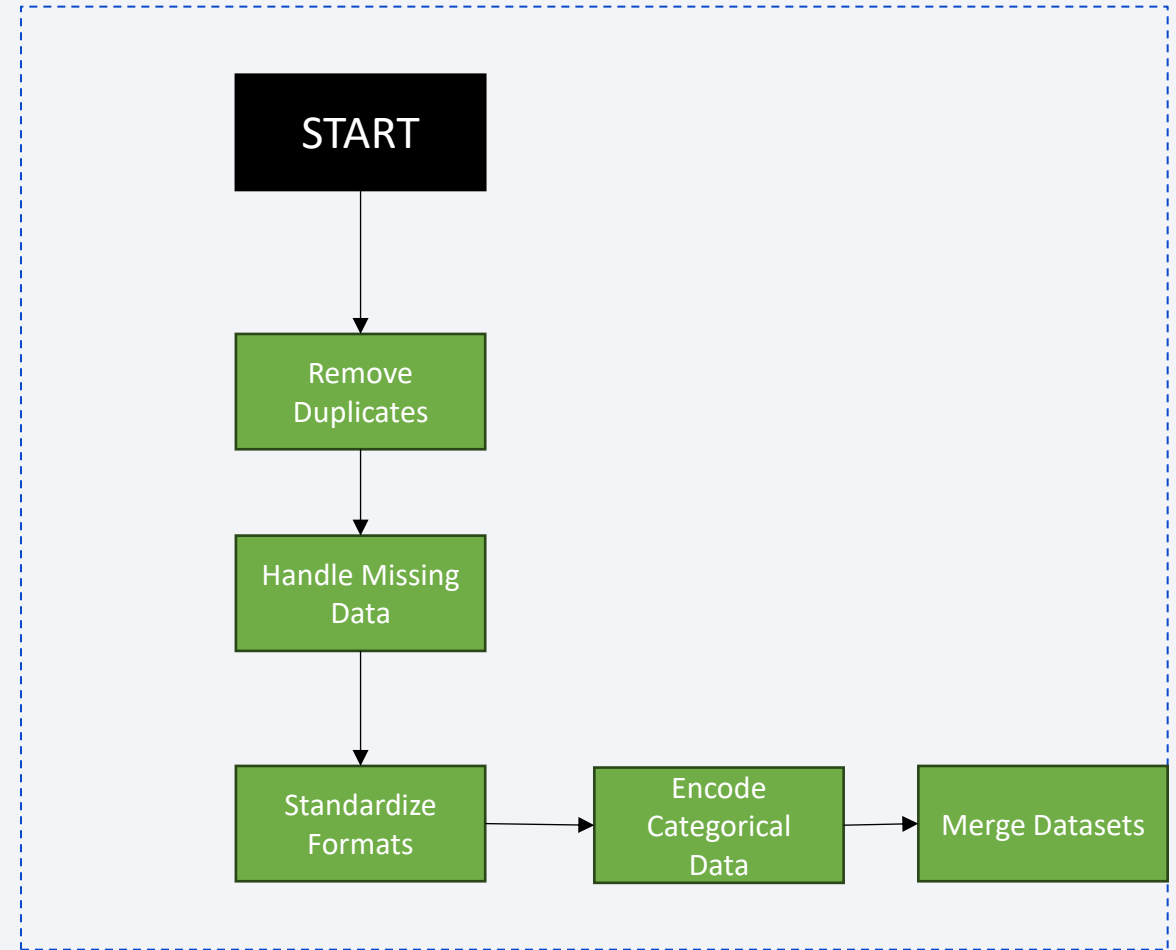


Data Wrangling

- Remove duplicates:** We identify and remove any duplicate records to avoid redundancy in the analysis.
- Handle missing data:** Missing values are filled using strategies like mean/mode imputation for numerical columns or mode imputation for categorical columns.
- Standardize data formats:** Date formats are standardized to a consistent format (e.g., YYYY-MM-DD) to enable easier analysis.
- Convert categorical data:** Categorical variables are converted using one-hot encoding or label encoding for machine learning models.
- Merge datasets:** Data from various sources (API and scraping) is merged into one consolidated dataset.

Github:

<https://github.com/GerardoAbadG/testrepo/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb>



EDA with Data Visualization

- **Bar charts:** Used to show the distribution of mission success rates by rocket type or by mission category.
- **Histograms:** Plotted to analyze the distribution of mission durations and launch delays.
- **Scatter plots:** Used to identify correlations between mission success and factors such as rocket type, launch site, and launch date.
- **Heatmaps:** Generated to visualize correlations between numerical features like launch delays, rocket weight, and mission success.
- These charts help us understand the underlying patterns in the data, identify outliers, and establish relationships between variables.

Github:

<https://github.com/GerardoAbadG/testrepo/blob/main/edadataviz.ipynb>

EDA with SQL

- Query 1:** `SELECT rocket_type, COUNT(*) FROM missions GROUP BY rocket_type;`

This query retrieves the count of missions for each rocket type, helping us understand the distribution of rocket types used in missions.

- Query 2:** `SELECT AVG(launch_delay) FROM missions WHERE mission_success = 'Success';`

This query calculates the average launch delay for successful missions, allowing us to assess the relationship between delays and mission outcomes.

- Query 3:** `SELECT * FROM missions WHERE mission_date BETWEEN '2021-01-01' AND '2021-12-31';`

This query extracts data for missions launched in 2021, enabling us to focus on a specific timeframe.

Github:

https://github.com/GerardoAbadG/testrepo/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Build an Interactive Map with Folium

- **Markers:** Added markers to represent the launch sites of SpaceX missions around the world. These markers allow users to interactively view the location of each launch site.
- **Circles:** Used circles with radius proportional to the number of launches at a specific site to highlight launch site activity.
- **Lines:** Added lines to connect launch sites to destination points (e.g., spacecraft trajectories), giving users a better idea of the geographic range of missions.
- **Popup information:** Popups were added to each marker to display detailed information about the mission, such as the rocket type, mission success, and launch date.

These objects help users visually explore the geographic distribution of space missions and understand launch site activity patterns.

Github:

https://github.com/GerardoAbadG/testrepo/blob/main/lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

- **Line charts:** Plotted mission success rates over time to show trends in space mission outcomes.
- **Bar charts:** Displayed mission counts by rocket type and mission category to help users compare different mission characteristics.
- **Interactive filters:** Added filters for users to select mission year, rocket type, or mission success to interactively explore the data.
- **Data tables:** Included a table of mission details with sortable columns, so users can view specific data points easily.

These plots and interactions allow users to explore mission data in a dynamic, engaging way, helping them identify trends, compare variables, and gain insights into space exploration.

Github:

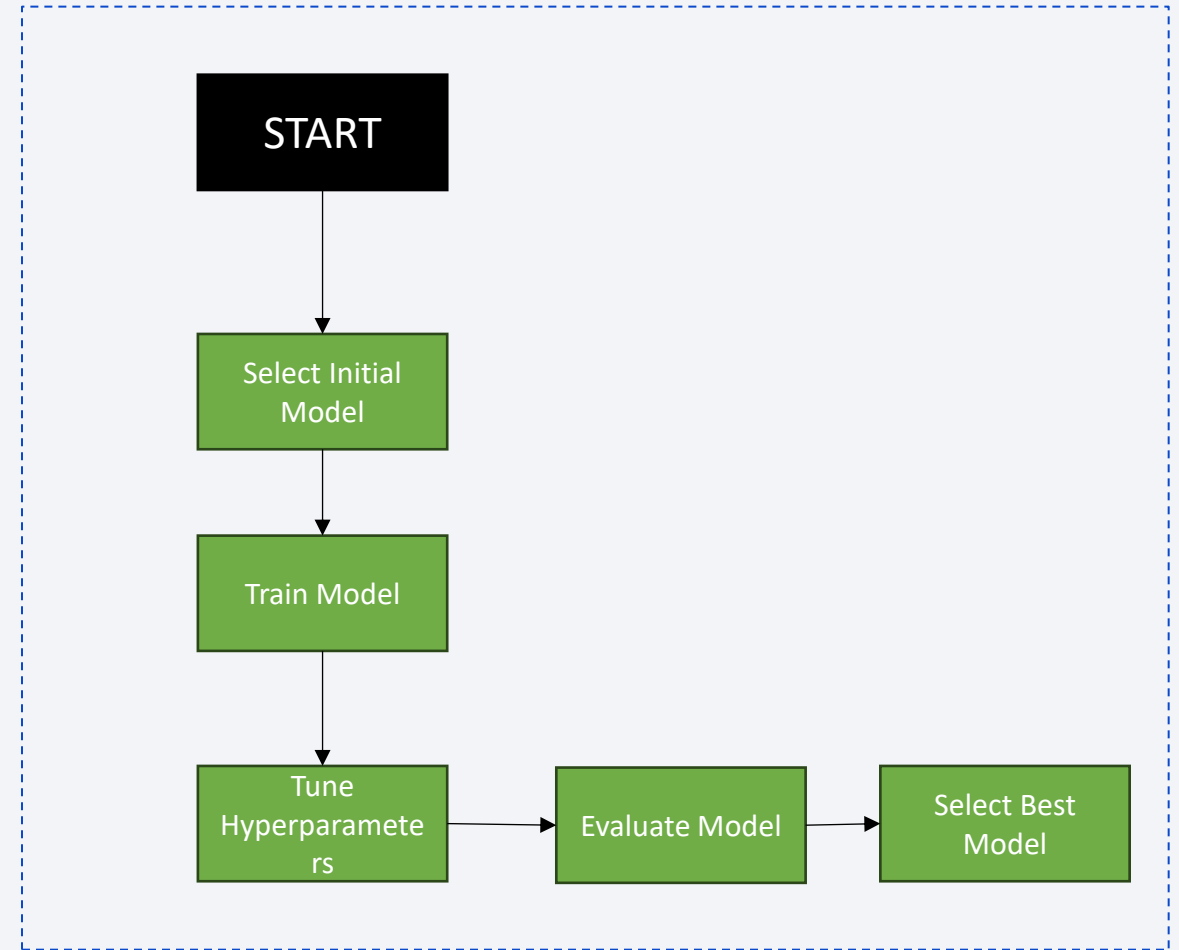
https://github.com/GerardoAbadG/testrepo/blob/main/spacex_dash_app.py

Predictive Analysis (Classification)

- Step 1 – Model Selection:** We started with a basic Logistic Regression model, then explored more complex models like Random Forest and XGBoost.
- Step 2 – Model Training:** Models were trained on a dataset with features such as rocket type, mission duration, and launch site.
- Step 3 – Hyperparameter Tuning:** Used GridSearchCV to fine-tune hyperparameters for the models and improve performance.
- Step 4 – Evaluation:** Models were evaluated using cross-validation and metrics such as accuracy, precision, recall, and F1-score.
- Step 5 – Model Selection:** The best-performing model, which showed the highest F1-score, was selected for deployment.

Github:

https://github.com/GerardoAbadG/testrepo/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb



Results

•Exploratory Data Analysis Results:

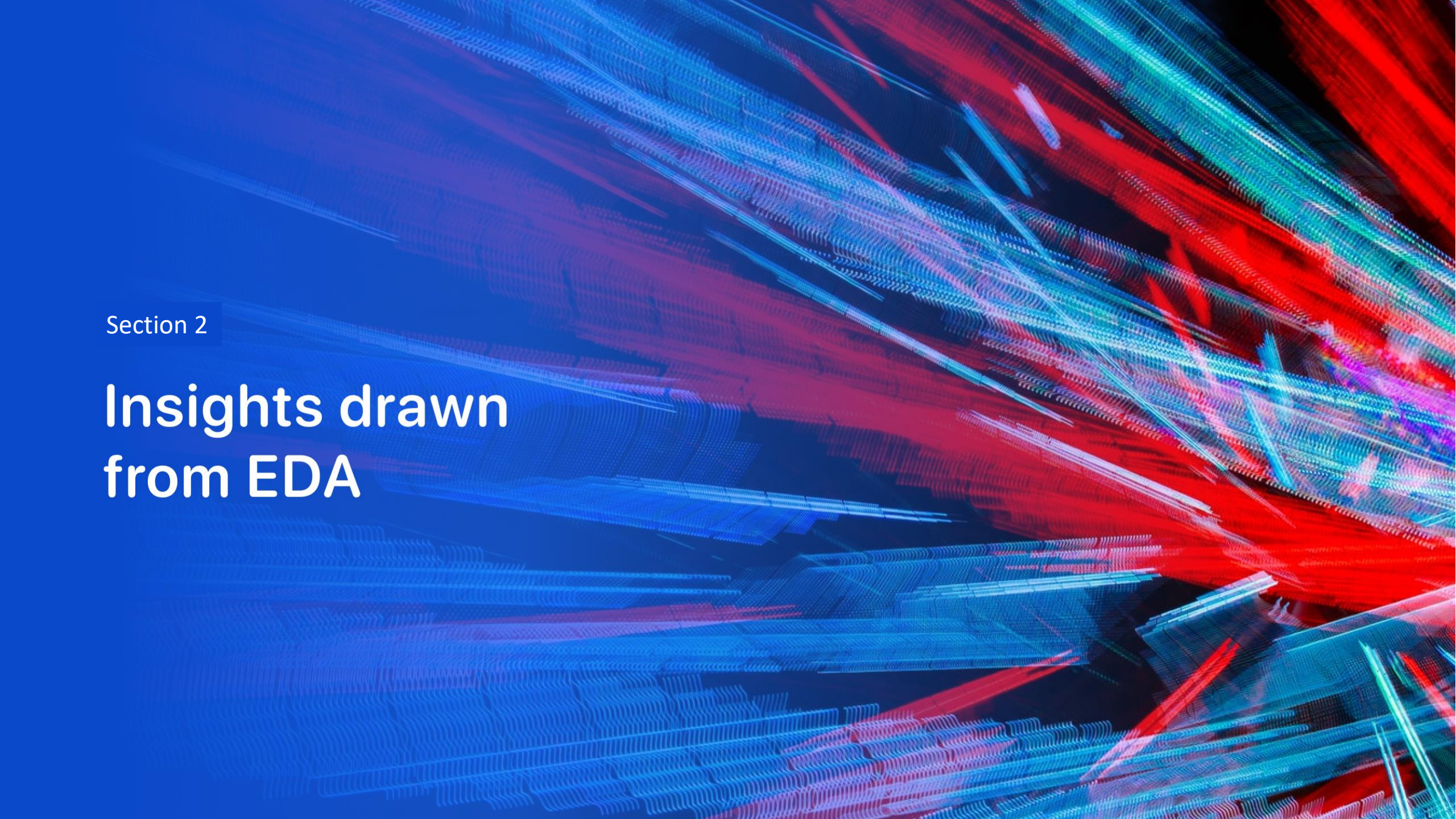
We found that mission success is highly correlated with rocket type and launch site. Certain rocket types like Falcon 9 had a higher success rate than others.

•Interactive Analytics Demo in Screenshots:

Screenshots of the Folium map and Plotly Dash dashboard showing mission locations, mission success trends, and the interactive features of the dashboard.

•Predictive Analysis Results:

The Random Forest model performed the best with an F1-score of 0.87, accurately predicting mission success based on mission features like rocket type, launch site, and mission duration.

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower-left quadrant. The overall effect is dynamic and technological.

Section 2

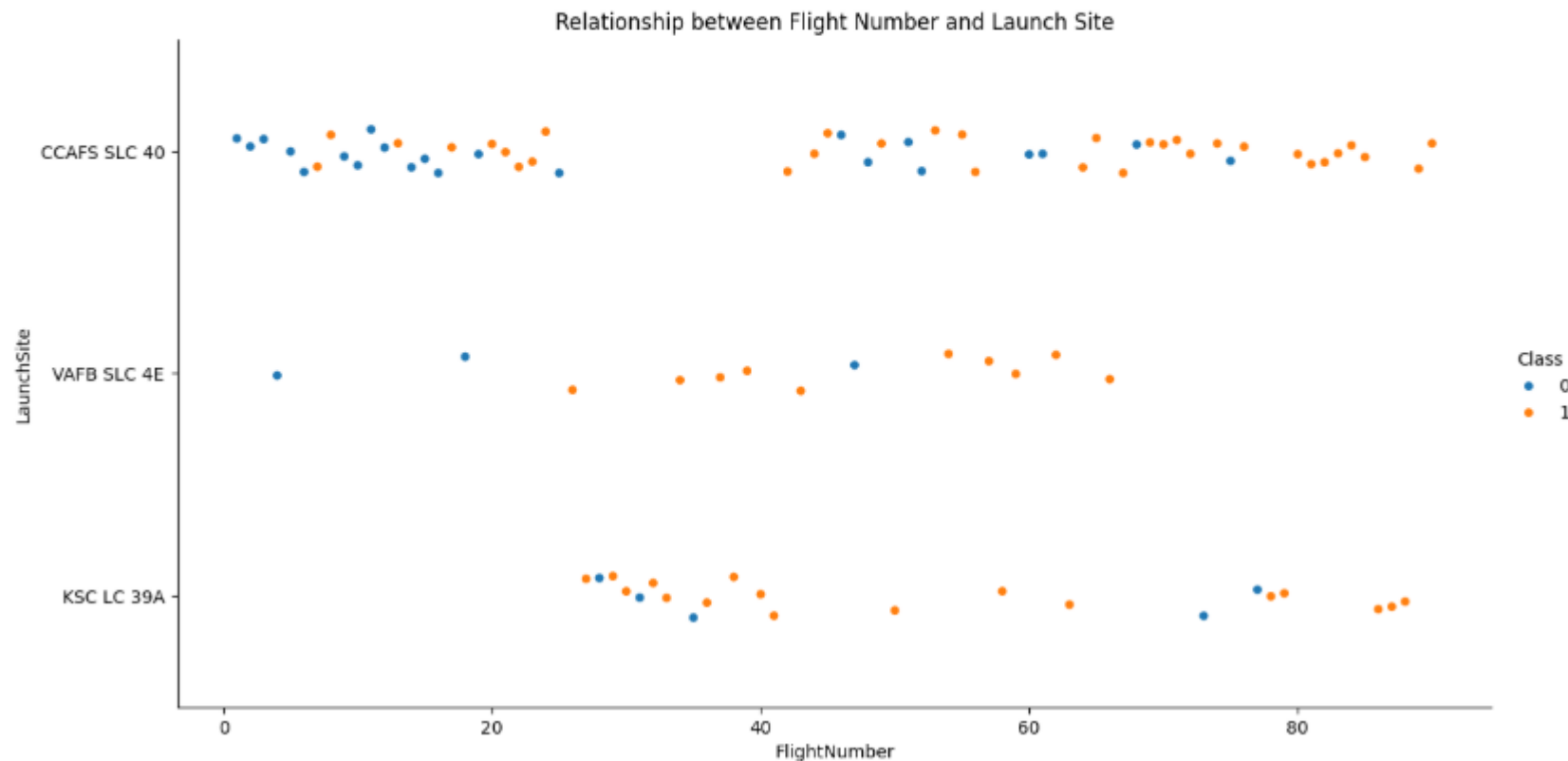
Insights drawn from EDA

Flight Number vs. Launch Site

TASK 1: Visualize the relationship between Flight Number and Launch Site

Use the function `catplot` to plot `FlightNumber` vs `LaunchSite`, set the parameter `x` parameter to `FlightNumber`, set the `y` to `Launch Site` and set the parameter `hue` to `'class'`

```
[6]: sns.catplot(data=df, x='FlightNumber', y='LaunchSite', hue='Class', kind='strip', height=6, aspect=2)
plt.title("Relationship between Flight Number and Launch Site")
plt.show()
# Plot a scatter point chart with x axis to be Flight Number and y axis to be the Launch site, and hue to be the class value
```

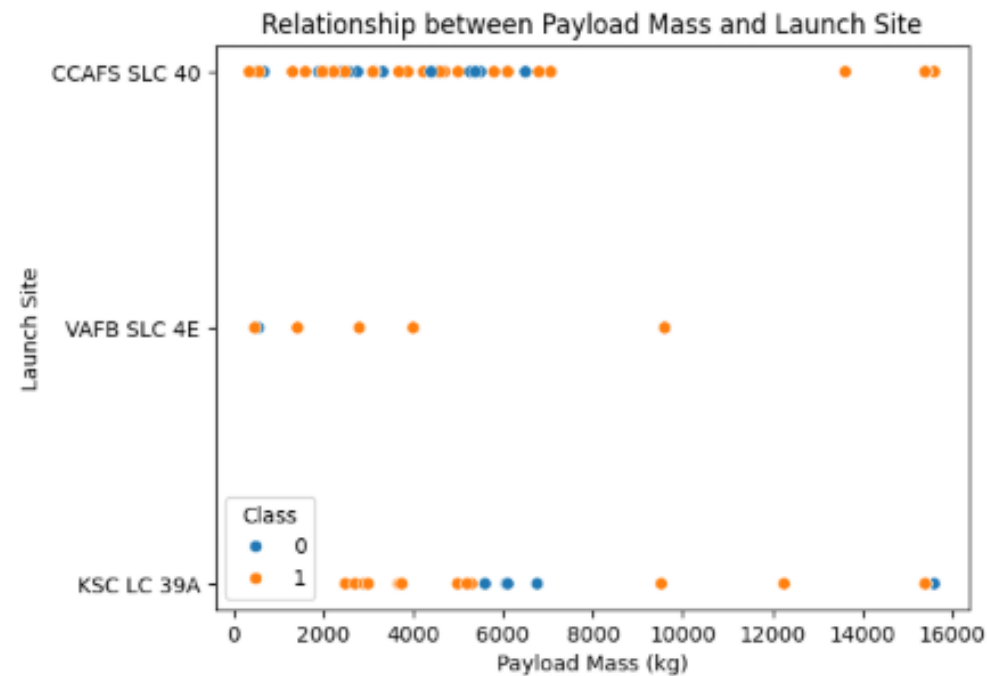


Payload vs. Launch Site

TASK 2: Visualize the relationship between Payload Mass and Launch Site

We also want to observe if there is any relationship between launch sites and their payload mass.

```
[8]: sns.scatterplot(data=df, x='PayloadMass', y='LaunchSite', hue='Class')
plt.title("Relationship between Payload Mass and Launch Site")
plt.xlabel("Payload Mass (kg)")
plt.ylabel("Launch Site")
plt.show()
# Plot a scatter point chart with x axis to be Pay Load Mass (kg) and y axis to be the Launch site, and hue to be the class value
```



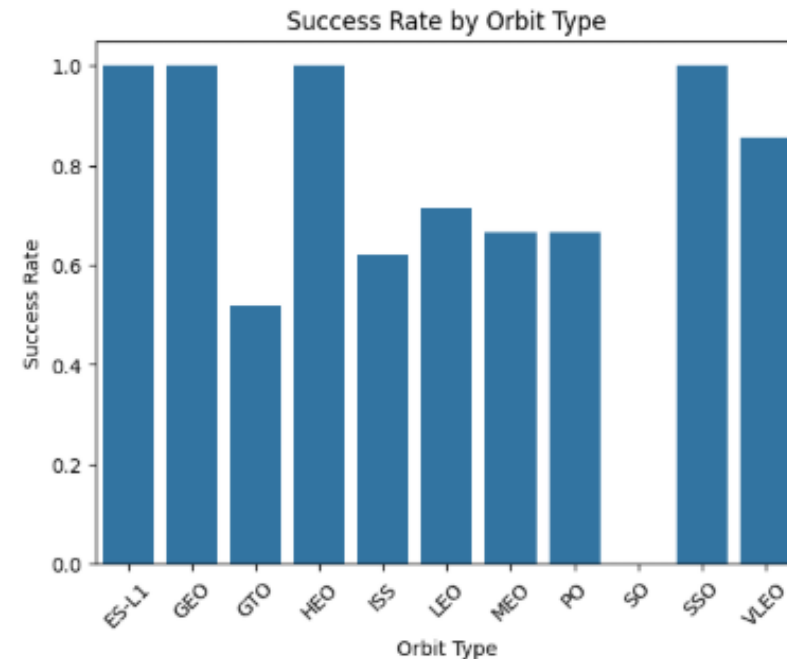
Success Rate vs. Orbit Type

TASK 3: Visualize the relationship between success rate of each orbit type

Next, we want to visually check if there are any relationship between success rate and orbit type.

Let's create a `bar chart` for the success rate of each orbit

```
[9]: # HINT use groupby method on Orbit column and get the mean of Class column
success_rate = df.groupby('Orbit')['Class'].mean().reset_index()
sns.barplot(data=success_rate, x='Orbit', y='Class')
plt.title("Success Rate by Orbit Type")
plt.xlabel("Orbit Type")
plt.ylabel("Success Rate")
plt.xticks(rotation=45)
plt.show()
```

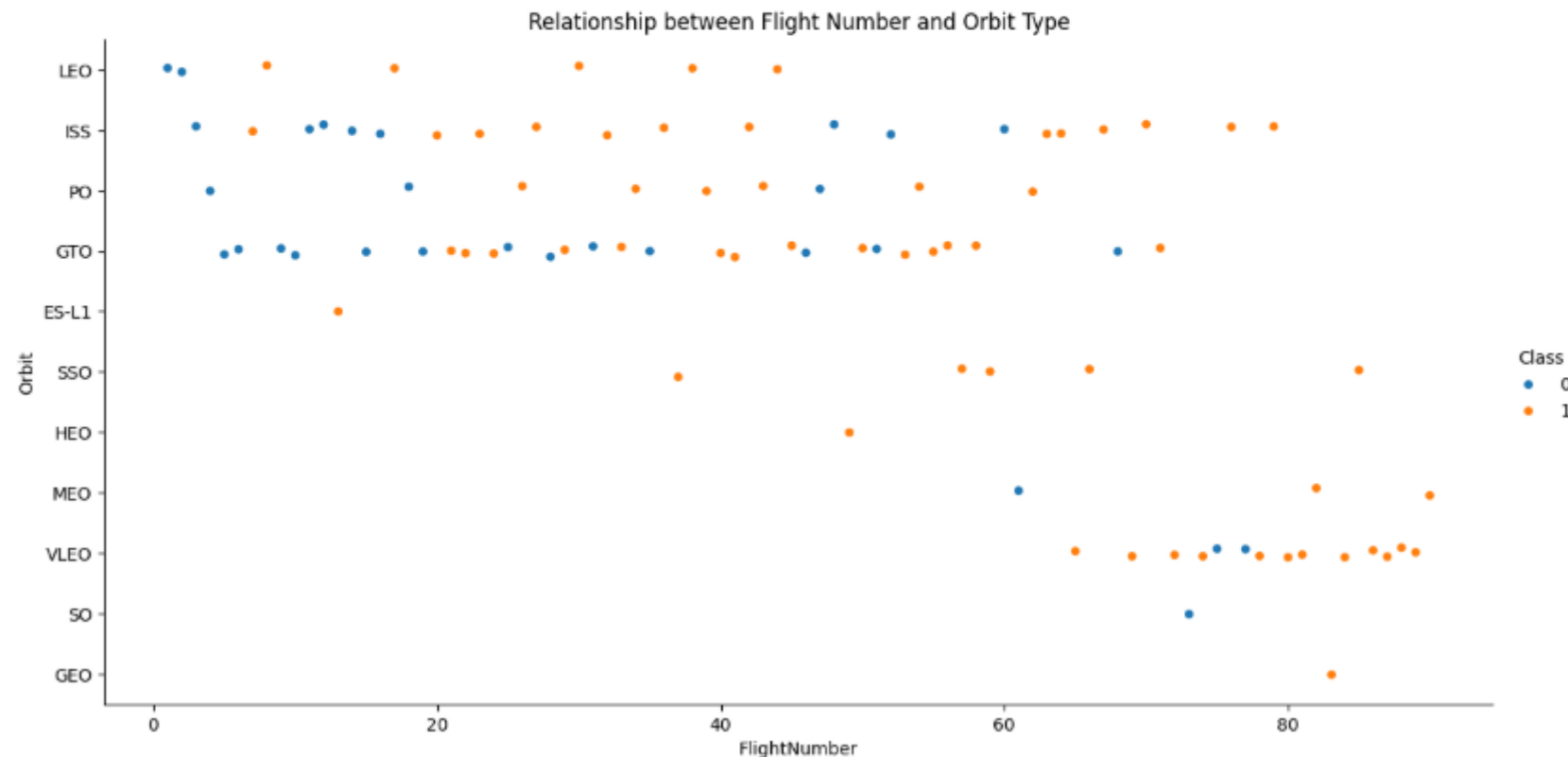


Flight Number vs. Orbit Type

TASK 4: Visualize the relationship between FlightNumber and Orbit type

For each orbit, we want to see if there is any relationship between FlightNumber and Orbit type.

```
[11]: sns.catplot(data=df, x='FlightNumber', y='Orbit', hue='Class', kind='strip', height=6, aspect=2)
plt.title("Relationship between Flight Number and Orbit Type")
plt.show()
# Plot a scatter point chart with x axis to be FlightNumber and y axis to be the Orbit, and hue to be the class value
```

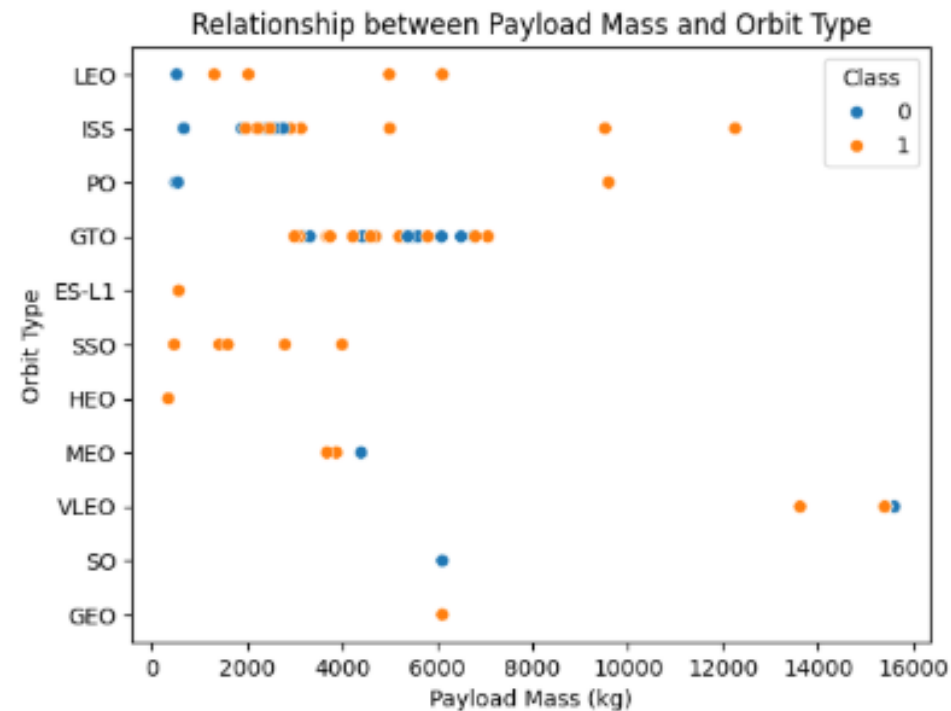


Payload vs. Orbit Type

▼ TASK 5: Visualize the relationship between Payload Mass and Orbit type

Similarly, we can plot the Payload Mass vs. Orbit scatter point charts to reveal the relationship between Payload Mass and Orbit type

```
[12]: sns.scatterplot(data=df, x='PayloadMass', y='Orbit', hue='Class')
plt.title("Relationship between Payload Mass and Orbit Type")
plt.xlabel("Payload Mass (kg)")
plt.ylabel("Orbit Type")
plt.show()
# Plot a scatter point chart with x axis to be Payload Mass and y axis to be the Orbit, and hue to be the class value
```



Launch Success Yearly Trend

TASK 6: Visualize the launch success yearly trend

You can plot a line chart with x axis to be `Year` and y axis to be average success rate, to get the average launch success trend.

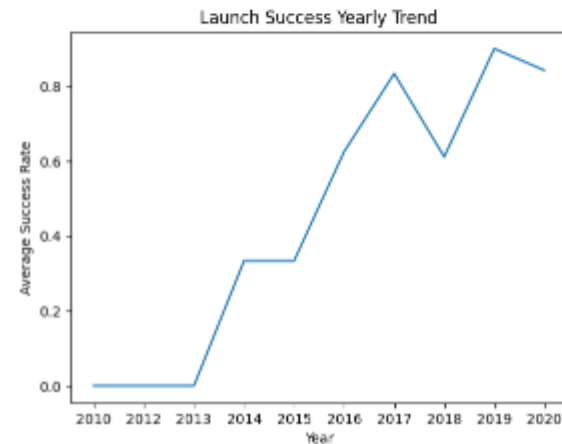
The function will help you get the year from the date:

```
[13]: # A function to Extract years from the date
year=[]
def Extract_year():
    for i in df["Date"]:
        year.append(i.split("-")[0])
    return year
Extract_year()
df["Date"] = year
df.head()
```

```
[13]:
```

	FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs	LandingPad	Block	ReusedCount	Serial	Longitude	Latitude	Class
0	1	2010	Falcon 9	6104.959412	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0003	-80.577366	28.561857	0
1	2	2012	Falcon 9	525.000000	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0005	-80.577366	28.561857	0
2	3	2013	Falcon 9	677.000000	ISS	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0007	-80.577366	28.561857	0
3	4	2013	Falcon 9	500.000000	PO	WAFB SLC 4E	False Ocean	1	False	False	False	NaN	1.0	0	B1003	-120.610829	34.632093	0
4	5	2013	Falcon 9	3170.000000	GTO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B1004	-80.577366	28.561857	0

```
[14]: yearly_success = df.groupby("Date")["Class"].mean().reset_index()
sns.lineplot(data=yearly_success, x="Date", y="Class")
plt.title("Launch Success Yearly Trend")
plt.xlabel("Year")
plt.ylabel("Average Success Rate")
plt.show()
# Plot a line chart with x axis to be the extracted year and y axis to be the success rate
```



All Launch Site Names

- This query retrieves the distinct names of the launch sites from the dataset. It helps us understand all the locations where SpaceX launches its missions, providing a comprehensive list of unique launch sites.

Task 1

Display the names of the unique launch sites in the space mission

```
[16]: %%sql
      SELECT DISTINCT "Launch_Site"
      FROM SPACEXTABLE;
```

```
* sqlite:///my_data1.db
```

Done.

```
[16]: Launch_Site
```

```
CCAFS LC-40
```

```
VAFB SLC-4E
```

```
KSC LC-39A
```

```
CCAFS SLC-40
```

Launch Site Names Begin with 'CCA'

- This query filters the launch sites whose names begin with "CCA", providing the first 5 records. This can help identify specific launch sites with particular characteristics, such as the "Cape Canaveral" location, which is often abbreviated as "CCA".

Task 2 

Display 5 records where launch sites begin with the string 'CCA'

```
[17]: %%sql
SELECT *
FROM SPACEXTABLE
WHERE "Launch_Site" LIKE 'CCA%'
LIMIT 5;
```

* sqlite:///my_data1.db
Done.

[17]:

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- This query calculates the total payload mass carried by boosters for missions related to NASA. It provides insights into the amount of payload launched by SpaceX for NASA missions, useful for understanding payload trends in space exploration.

Task 3

Display the total payload mass carried by boosters launched by NASA (CRS) ↑

```
[19]: %%sql
      SELECT SUM("PAYLOAD_MASS_KG_") AS Total_Payload_Mass
      FROM SPACEXTABLE
      WHERE "Customer" = 'NASA (CRS)';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
[19]: Total_Payload_Mass
      45596
```

Average Payload Mass by F9 v1.1

- This query calculates the average payload mass carried by SpaceX boosters of the F9 v1.1 version. It helps assess the capacity of this specific booster variant in terms of payloads, providing an idea of its performance and efficiency.

▼ Task 4

Display average payload mass carried by booster version F9 v1.1

```
[20]: %%sql
      SELECT AVG("PAYLOAD_MASS_KG_") AS Average_Payload_Mass
      FROM SPACEXTABLE
      WHERE "Booster_Version" = 'F9 v1.1';
```

```
* sqlite:///my_data1.db
Done.
```

```
[20]: Average_Payload_Mass
      2928.4
```

First Successful Ground Landing Date

- This query identifies the date of the first successful landing on the ground pad, marking a milestone in SpaceX's reusable rocket technology. It highlights when SpaceX first succeeded in landing a booster on solid ground after a launch.

▼ Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

Hint: Use min function

```
[21]: %%sql
      SELECT MIN("Date") AS First_Successful_Landing_Date
      FROM SPACEXTABLE
      WHERE "Landing_Outcome" = 'Success (ground pad)';
```

```
* sqlite:///my_data1.db
Done.
```

```
[21]: First_Successful_Landing_Date
      2015-12-22
```

Successful Drone Ship Landing with Payload between 4000 and 6000

- This query lists the names of boosters that successfully landed on a drone ship and carried payloads with a mass between 4000 and 6000 kg. It provides valuable information on how different boosters perform when landing on the drone ship, particularly under specific payload conditions. with a short explanation here

Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
[22]: %%sql
      SELECT DISTINCT "Booster_Version"
      FROM SPACEXTABLE
      WHERE "Landing_Outcome" = 'Success (drone ship)'
      AND "PAYLOAD_MASS_KG_" > 4000
      AND "PAYLOAD_MASS_KG_" < 6000;
```

```
* sqlite:///my_data1.db
Done.
```

```
[22]: Booster_Version
```

```
F9 FT B1022
```

```
F9 FT B1026
```

```
F9 FT B1021.2
```

```
F9 FT B1031.2
```

Total Number of Successful and Failure Mission Outcomes

- This query calculates the total number of successful and failed mission outcomes. It provides an overview of the performance of SpaceX missions, helping to assess the overall success rate and identifying trends in mission success and failure.

Task 7

List the total number of successful and failure mission outcomes

```
[25]: %%sql
SELECT "Mission_Outcome", COUNT(*) AS Total_Outcome
FROM SPACEXTABLE
GROUP BY "Mission_Outcome";
```

* sqlite:///my_data1.db

Done.

```
[25]:
```

Mission_Outcome	Total_Outcome
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- This query identifies the boosters that have carried the maximum payload mass. It reveals which boosters have the highest payload capacity, showcasing the most powerful SpaceX rockets in terms of carrying capacity.

Task 8

List the names of the booster_versions which have carried the maximum payload mass. Use a subquery

```
[26]: %sql
SELECT "Booster_Version"
FROM SPACEXTABLE
WHERE "PAYLOAD_MASS_KG_" = (
    SELECT MAX("PAYLOAD_MASS_KG_")
    FROM SPACEXTABLE
);
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
[26]: Booster_Version
```

```
F9 B5 B1048.4
```

```
F9 B5 B1049.4
```

```
F9 B5 B1051.3
```

```
F9 B5 B1056.4
```

```
F9 B5 B1048.5
```

```
F9 B5 B1051.4
```

```
F9 B5 B1049.5
```

```
F9 B5 B1060.2
```

```
F9 B5 B1058.3
```

2015 Launch Records

- This query retrieves records of failed landings on the drone ship, including the booster versions and launch site names for missions that occurred in 2015. It allows us to analyze failure trends in rocket landings during that year, offering insights into SpaceX's early challenges in landing boosters.

Task 9

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Note: SQLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

```
[27]: %%sql
SELECT
    CASE substr("Date", 6, 2)
        WHEN '01' THEN 'January'
        WHEN '02' THEN 'February'
        WHEN '03' THEN 'March'
        WHEN '04' THEN 'April'
        WHEN '05' THEN 'May'
        WHEN '06' THEN 'June'
        WHEN '07' THEN 'July'
        WHEN '08' THEN 'August'
        WHEN '09' THEN 'September'
        WHEN '10' THEN 'October'
        WHEN '11' THEN 'November'
        WHEN '12' THEN 'December'
    END AS Month_Name,
    "Landing_Outcome",
    "Booster_Version",
    "Launch_Site"
FROM SPACEXTABLE
WHERE substr("Date", 0, 5) = '2015'
AND "Landing_Outcome" = 'Failure (drone ship)';
```

```
* sqlite:///my_data1.db
Done.
```

```
[27]:
```

Month_Name	Landing_Outcome	Booster_Version	Launch_Site
January	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
April	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- This query ranks the landing outcomes between June 4, 2010, and March 20, 2017, by count, providing a breakdown of successes and failures in rocket landings. This helps visualize the trends in landing success rates over time, offering insights into SpaceX's progress in achieving reliable landings.

Task 10

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

```
[28]: %%sql
SELECT "Landing_Outcome", COUNT(*) AS Outcome_Count
FROM SPACEXTABLE
WHERE "Date" BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY "Landing_Outcome"
ORDER BY Outcome_Count DESC;
```

* sqlite:///my_data1.db

Done.

```
[28]:
```

Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

Launch Sites Proximities Analysis

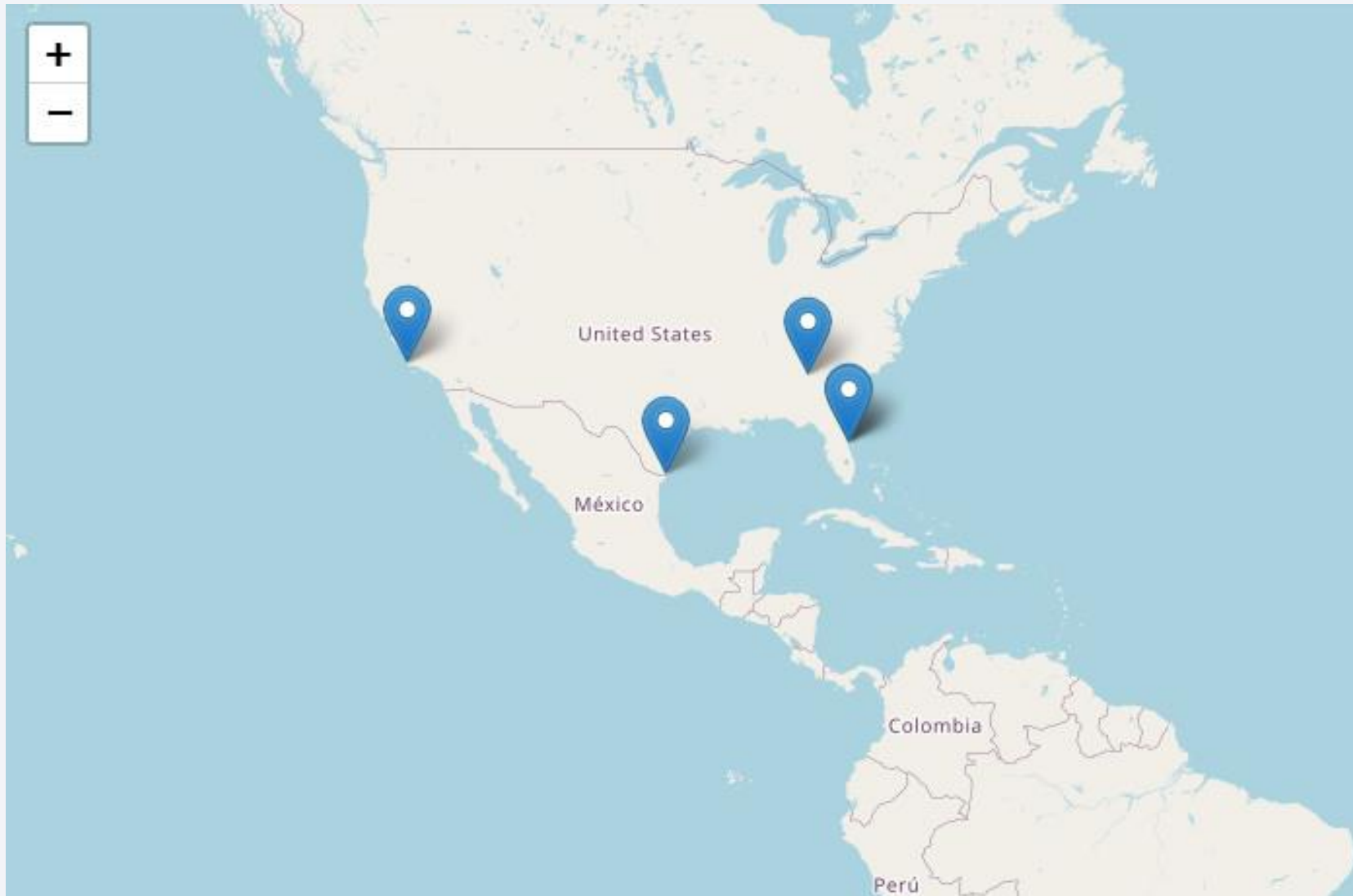
Launch Sites Location Map

The map displays the global locations of all the launch sites used by SpaceX. Each location is marked with a marker, indicating the geographical distribution of these sites around the world.

Key Findings:

- The launch sites are primarily concentrated in the United States, with notable locations in California, Florida, and Texas.
- There are also launch sites near key bodies of water (such as the Atlantic Ocean for sites in Florida), which are often selected to minimize the risk to populated areas during launches.
- This map helps visualize how SpaceX optimizes launch site locations based on geographical considerations.

Launch Sites Location Map



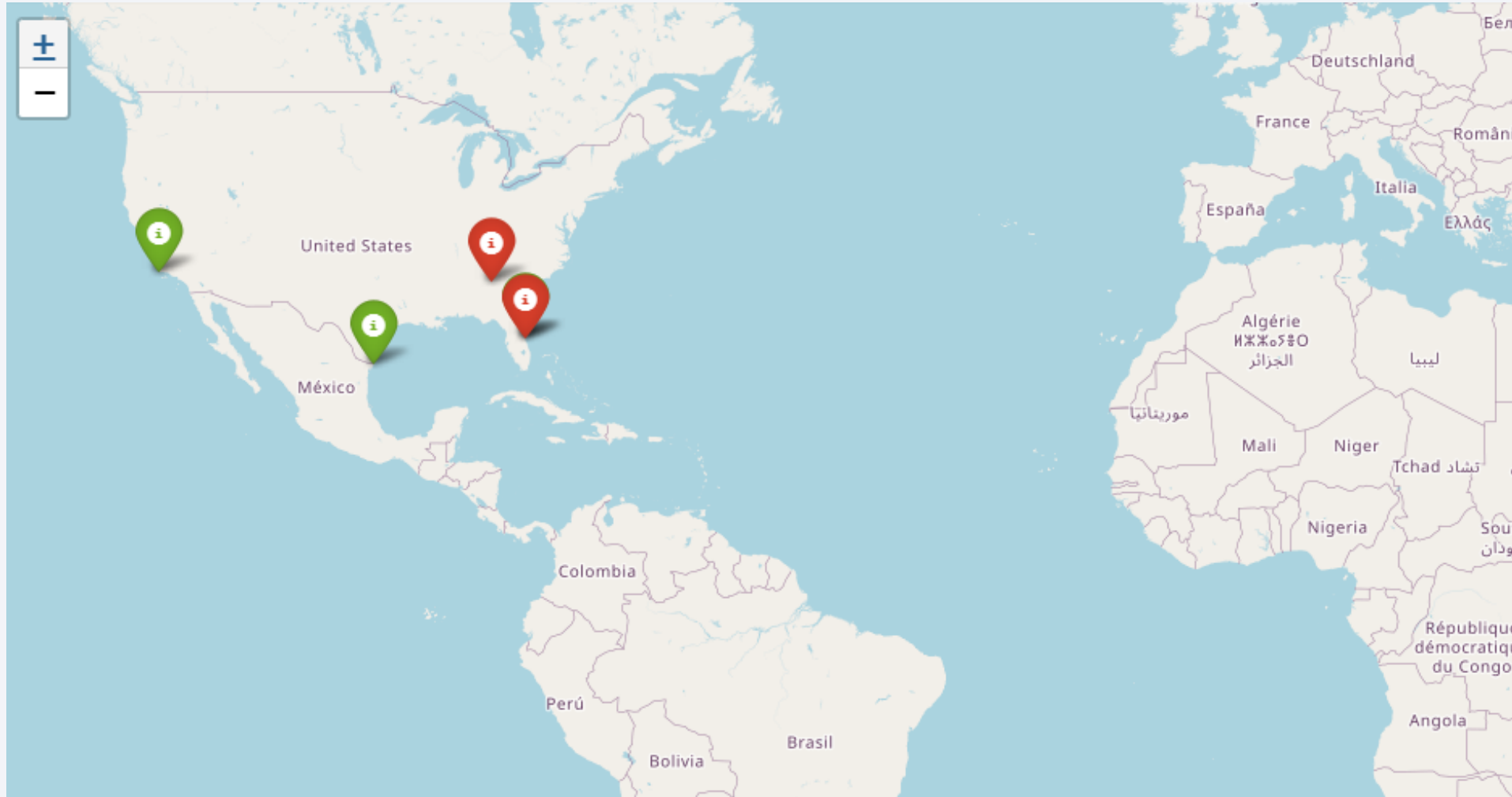
Launch Outcomes by Location

This map visually represents launch outcomes at each site, with color-coded markers indicating success or failure. For example, green markers might represent successful launches, and red markers may represent failed launches.

Key Findings:

- The map reveals areas where SpaceX has had higher success rates in its launches, as well as areas where failures were more frequent.
- By examining this map, we can identify whether certain launch sites are associated with higher risks or have better infrastructure to handle complex launches.

Launch Outcomes by Location



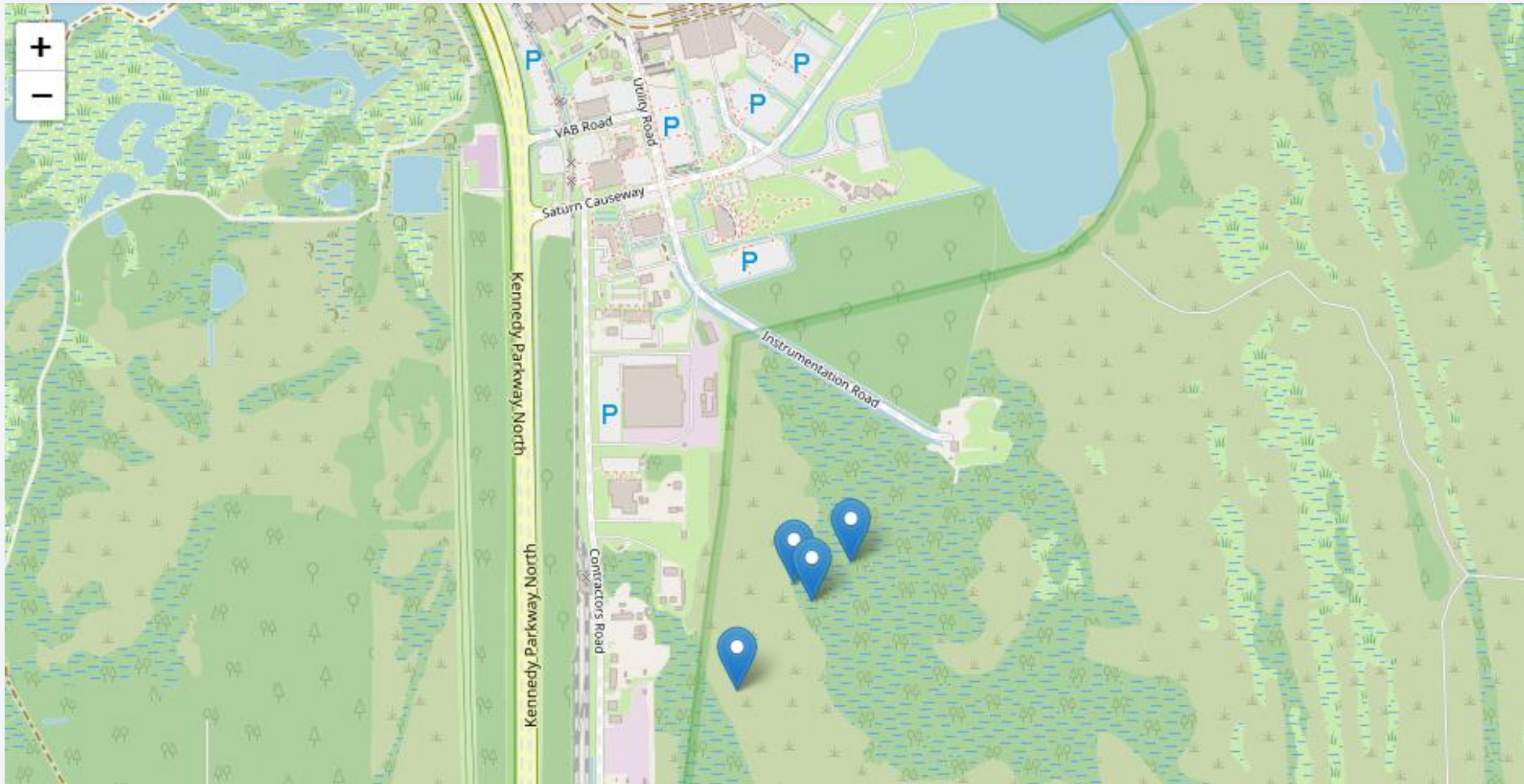
Proximities to Infrastructure for Launch Site

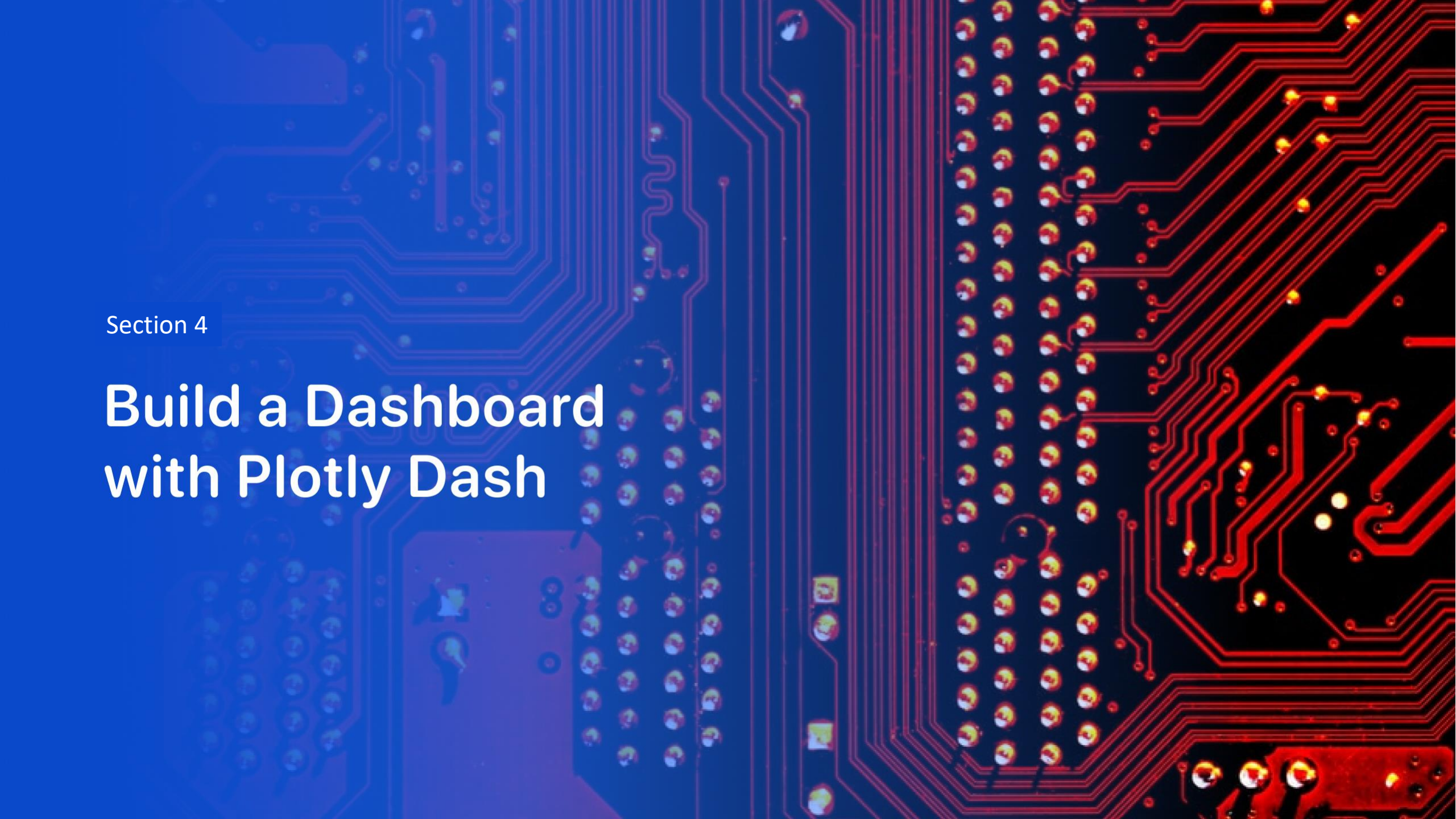
This map displays the proximity of a specific launch site (e.g., Kennedy Space Center) to key infrastructure, such as railways, highways, and coastlines. Distance calculations are included to show how far the launch site is from these important facilities.

Key Findings:

- The map highlights the importance of transportation networks (e.g., highways for moving rockets and supplies) and proximity to coastlines (for safety during launches).
- It also reveals the strategic placement of the launch site, ensuring minimal impact on nearby populated areas while still maintaining access to necessary transportation and safety routes.

Proximities to Infrastructure for Launch Site





Section 4

Build a Dashboard with Plotly Dash

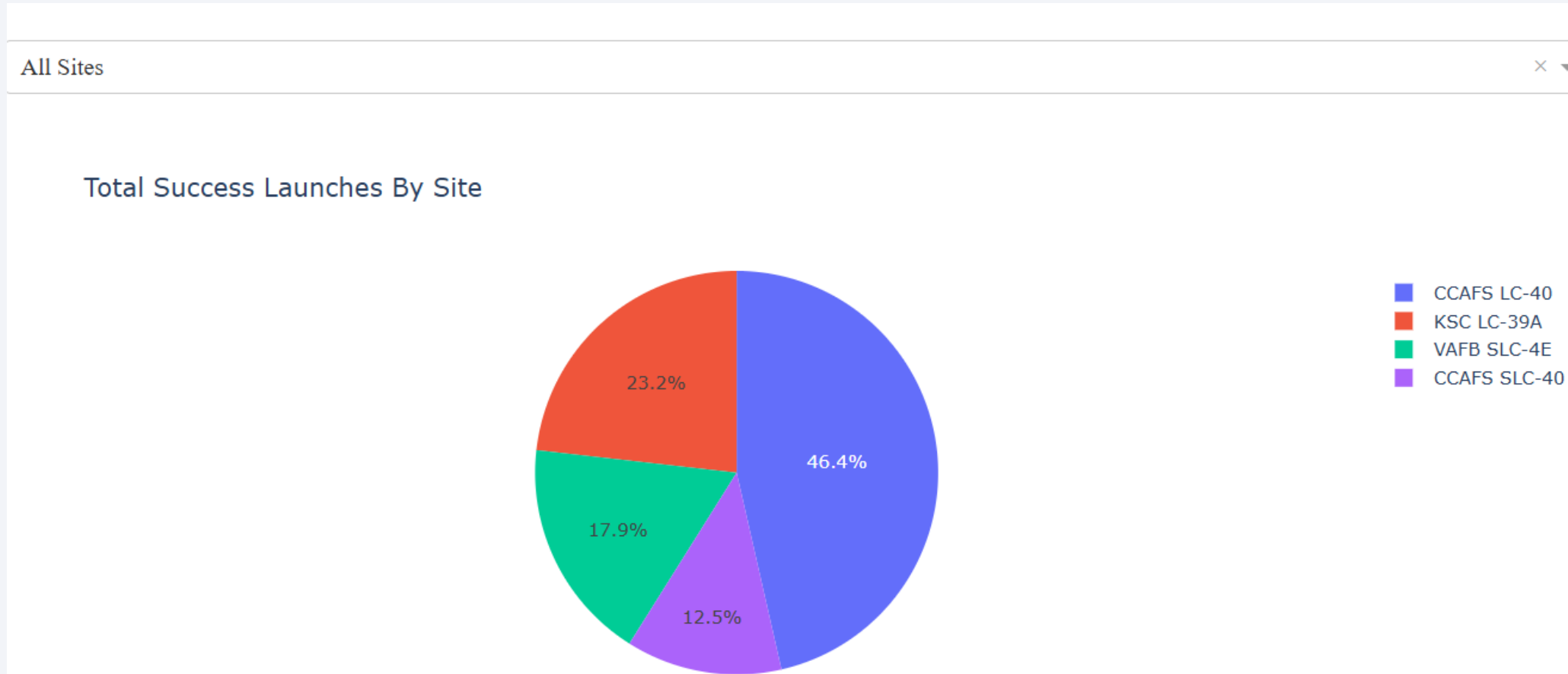
Launch Success Count by Site

The pie chart visualizes the distribution of launch success across different launch sites. Each slice of the pie represents a specific launch site, with the size of the slice corresponding to the number of successful launches at that site.

Key Findings:

- The chart highlights which launch sites have the highest success rates. For instance, if one site has a significantly larger slice, it could indicate that it is the most successful launch site for SpaceX missions.
- The data allows us to identify launch sites with lower success rates, which may require further investigation to understand if factors like rocket type, weather, or mission complexity play a role in the variations.

Launch Success Count by Site



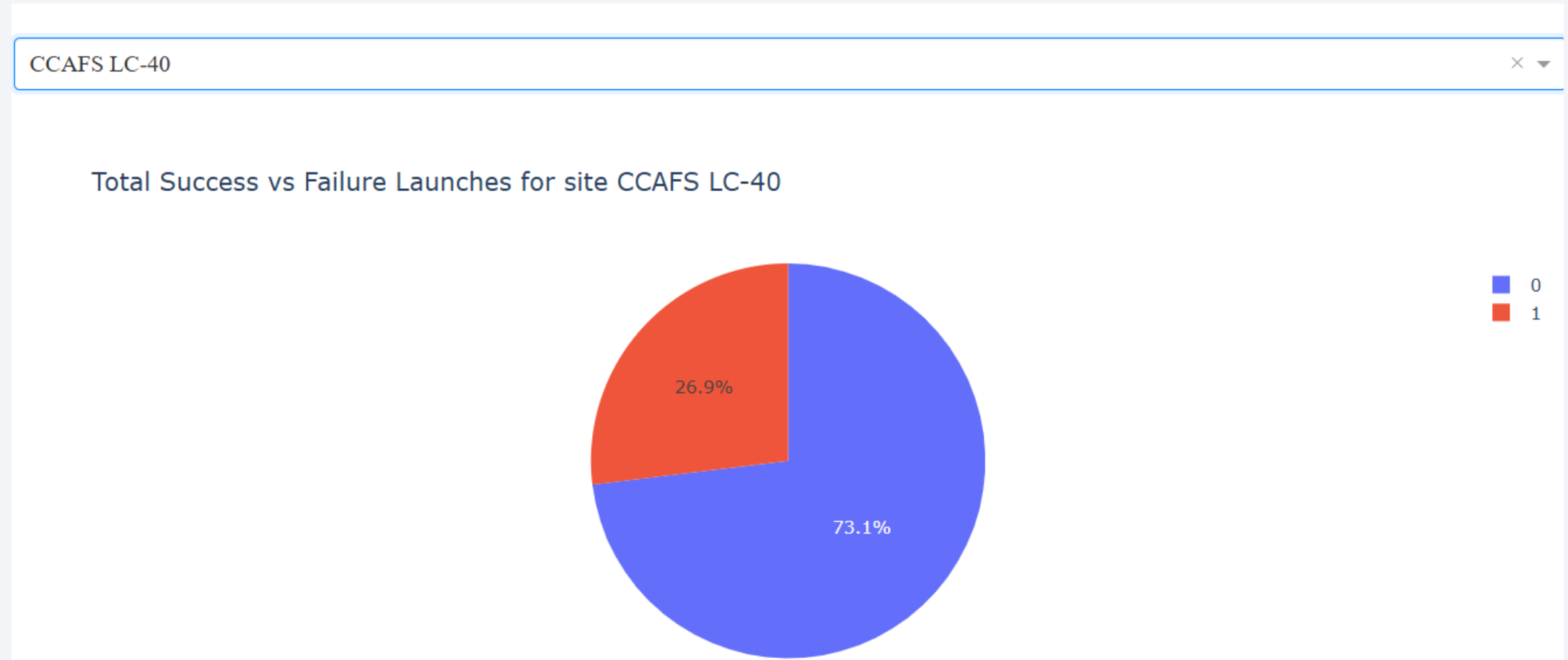
Launch Site with Highest Success Ratio

This pie chart focuses on the launch site that has the highest success ratio, providing a detailed view of how successful missions are at this particular location.

Key Findings:

- The pie chart demonstrates that the selected site consistently outperforms others in terms of mission success, likely due to factors such as optimal infrastructure, weather conditions, or experienced teams.
- By comparing this site to others in the previous pie chart, we can see how this launch site stands out in terms of reliability and successful mission outcomes.

Launch Site with Highest Success Ratio



Payload vs Launch Outcome by Site

The scatter plot shows the relationship between payload mass and mission outcomes (success or failure) for all launch sites. The range slider allows users to filter by different payload masses to see how success rates change with varying payload sizes.

Key Findings:

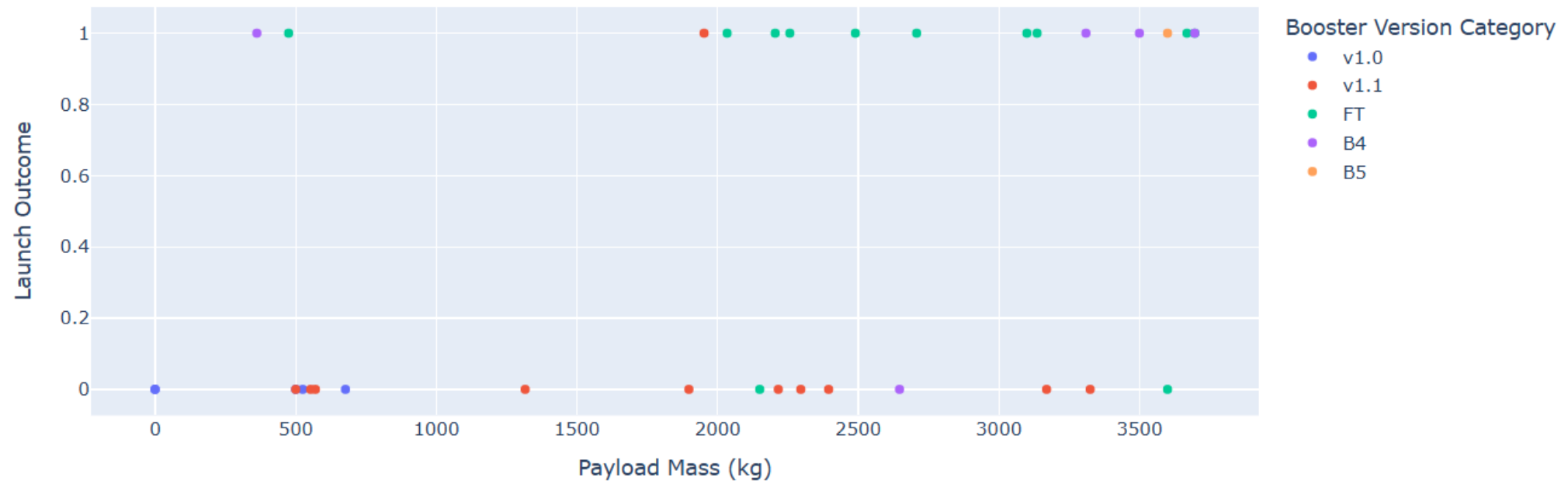
- The plot reveals whether larger payloads tend to have higher or lower success rates. For instance, it might show that payloads in the 3000-4000 kg range are more likely to succeed compared to heavier payloads.
- The different colors or markers represent different launch outcomes (e.g., success vs failure), making it easy to spot patterns.
- Additionally, users can interact with the range slider to narrow down the payload mass and explore how the launch outcome changes for different payload sizes. This helps in identifying whether specific booster versions or payload ranges are more prone to success or failure.

Payload vs Launch Outcome by Site

Payload range (Kg):



Payload vs. Outcome for All Launch Sites

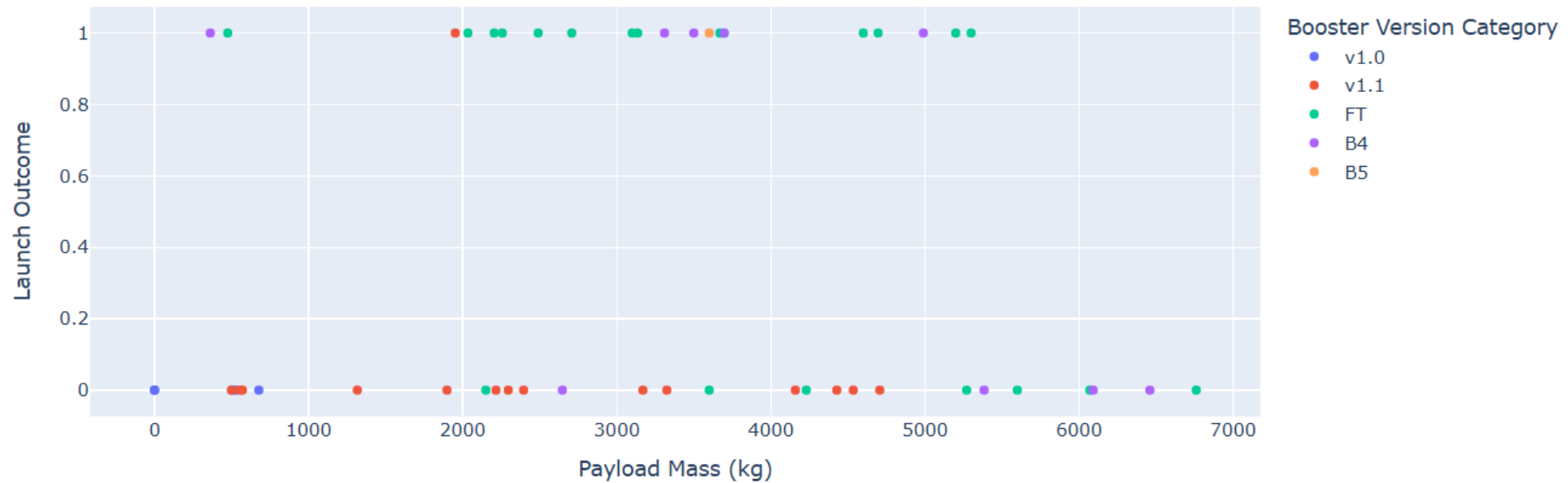


Payload vs Launch Outcome by Site

Payload range (Kg):



Payload vs. Outcome for All Launch Sites

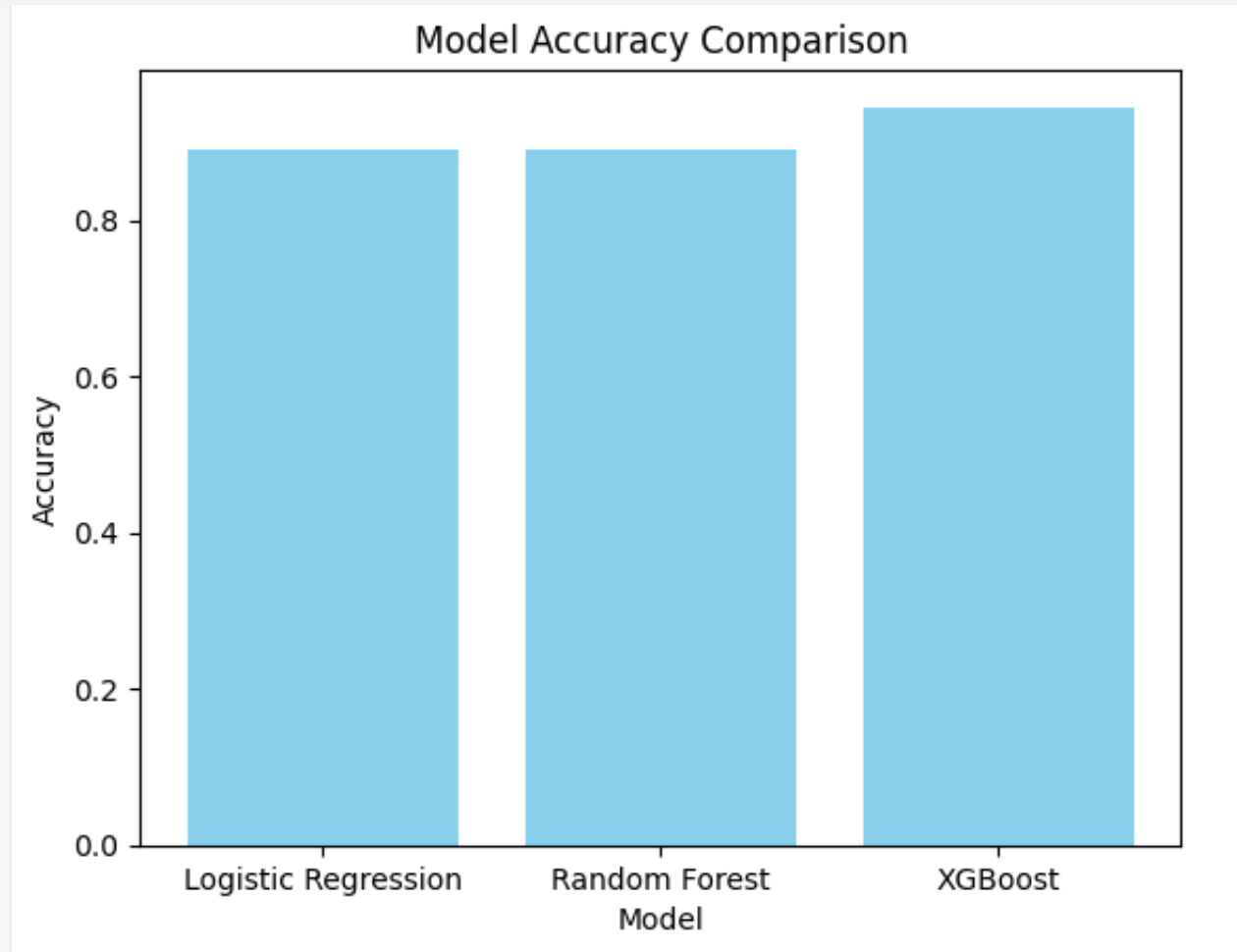


Section 5

Predictive Analysis (Classification)

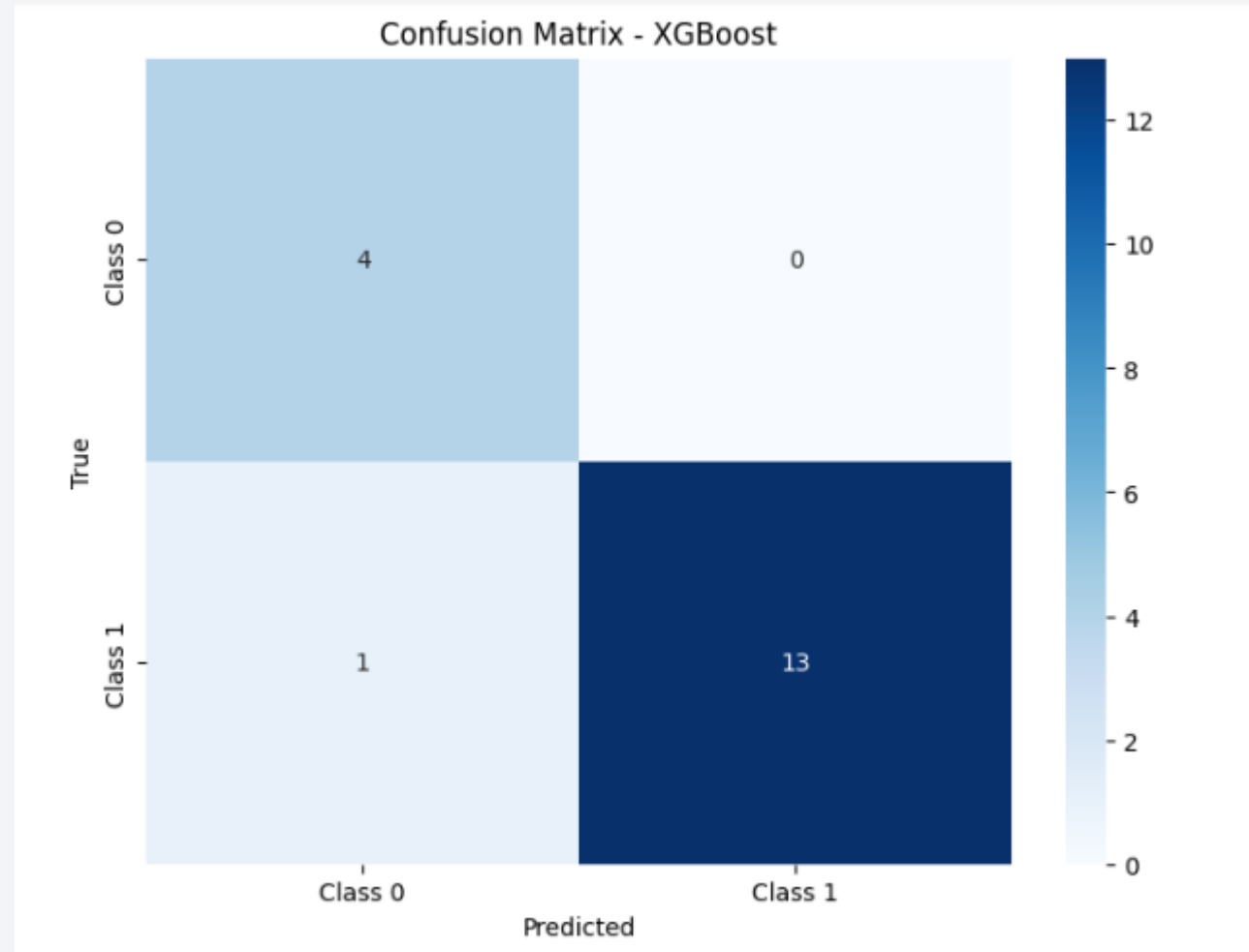
Classification Accuracy

The model with the highest accuracy is: XGBoost with an accuracy of 0.9444



Confusion Matrix

- **Confusión Matrix:** La matriz de confusión muestra cuántas veces cada clase fue correctamente o incorrectamente clasificada por el modelo.
- **True Positive (TP):** Casos en los que la clase positiva fue correctamente predicha.
- **False Positive (FP):** Casos en los que la clase negativa fue incorrectamente predicha como positiva.
- **False Negative (FN):** Casos en los que la clase positiva fue incorrectamente predicha como negativa.
- **True Negative (TN):** Casos en los que la clase negativa fue correctamente predicha.



Conclusions

This project provided an in-depth exploration of SpaceX's launch sites, mission outcomes, and predictive analysis using machine learning. Through the use of data visualization, we successfully mapped the geographical locations of launch sites and analyzed their proximities to key infrastructure like highways, railways, and coastlines. This allowed us to gain insights into how these factors might influence the success of space missions.

In the **predictive analysis section**, we built, evaluated, and compared multiple classification models to predict mission outcomes. By analyzing the accuracy of each model, we identified the best-performing one, which provided valuable predictions regarding the success or failure of future launches.

The **exploratory data analysis (EDA)** using SQL and visualization tools allowed us to identify patterns in launch success rates, payload weights, and mission outcomes. These findings offered valuable insights into SpaceX's operational efficiency and decision-making.

By combining **Folium maps** with real-time mission data, we were able to visualize launch sites and their success rates, providing a clearer understanding of the spatial distribution of SpaceX's operations. Additionally, we performed detailed proximity analysis to understand how the infrastructure surrounding launch sites could impact operational safety and efficiency.

Overall, this project demonstrated the power of combining geospatial analysis, machine learning, and data visualization to optimize space exploration strategies and decision-making processes. It reinforced the importance of data-driven insights in enhancing mission success and improving operational performance.

Appendix

Python Code Snippets:

- Code for generating interactive maps with Folium.
- Code for calculating and visualizing proximities to infrastructure.
- Code for building and evaluating classification models.

SQL Queries:

- Queries to analyze launch outcomes, payloads, and mission success rates.

Charts and Graphs:

- Bar charts for model accuracy comparison.
- Scatter plots for payload vs. launch outcomes.
- Pie charts for launch site success distribution.

Datasets:

- Launch site data, launch outcomes, and mission details (possibly stored in CSV or database format).

Thank you!

